Notes: Jeffers, Lyu, and Posenau (JFE, 2024)

The Risk and Return of Impact Investing Funds

Contents

1 Big picture

- **Question.** What are the risk exposures and risk-adjusted returns of private-market impact investing funds?
- **Why the question matters.** Impact funds pursue both financial and social goals. If they simply give up return for impact, they are concessionary. If they hedge risks investors dislike, they can be financially useful even with lower absolute returns.
- **Data contribution.** The paper creates a dataset of impact fund cash flows from the Impact Finance Research Consortium and Preqin.
- **Sample.** The main impact sample contains 94 funds with vintage years 1999–2015 and cash-flow data through 2021.
- **Benchmark design.** Each impact fund is compared to a matched private-market fund with similar asset class, vintage, and size, and also to a broader U.S. VC benchmark.
- **Methodological contribution.** The paper uses the gap between PME and GPME, the "PME wedge," to infer market risk exposure from cash flows alone.
- **Risk result.** Impact funds have lower market risk exposure than comparable private-market funds and VC funds.
- **Return result.** Impact funds underperform the public market on a market-risk-adjusted basis, but not necessarily more than comparable private-market funds.
- **Long-short result.** A portfolio long matched funds or VC funds and short impact funds has positive market risk exposure, confirming that benchmarks are more cyclical than impact.
- **Alternative risk models.** Adding a public sustainability factor helps explain impact fund returns more than it helps explain matched or VC funds. Emerging market exposure changes the interpretation for investors with different wealth portfolios.
- **Economic takeaway.** The financial merit of impact investing depends on the investor's risk model, wealth portfolio, and taste for impact. Impact investing may be concessionary relative to the public market, but it can also reduce investors' exposure to market risk relative to comparable private-market strategies.

2 Data and measurement

2.1 Impact fund data

- Impact fund cash flows are collected from the Impact Finance Research Consortium and Preqin.
- The IFRC component uses annual and quarterly financial statements from impact funds.
- The authors standardize contributions, distributions net of fees, and net asset value.
- The final impact sample contains 94 funds, vintages 1999–2015, with cash-flow data from 1999 to 2021.
- Funds have explicit financial and social or environmental goals.

Interpretation: The data contribution is central because private impact-fund cash flows are usually not publicly available. The paper therefore studies impact investing using fund-level cash flows rather than self-reported aggregate performance.

2.2 Mission categories

- Impact funds are classified by mission focus using sustainability reports, prospectuses, and websites.
- The broad categories are environmental, social, and social/environmental.
- Primary mission categories include clean technology, economic development, financial inclusion, food and agriculture, social services, and sustainable practices.

Interpretation: These classifications show that the sample is not only a clean-tech sample. It spans economic development, finance, social services, agriculture, and environmental goals.

2.3 Benchmark funds

- The first benchmark is a set of Preqin private-market funds matched to impact funds by asset class, vintage, and size.
- The second benchmark is a U.S.-focused VC sample from Preqin.
- Matched funds isolate the impact component while holding private-market characteristics as fixed as possible.
- VC funds provide a broad asset-class benchmark because many impact funds resemble VC or growth funds.

Interpretation: The matched benchmark answers: what if the investor invested in a similar private-market fund that did not have an impact mandate?

2.4 Private-market performance measures

The paper uses cash-flow-based measures, including:

$$Multiple = \frac{\sum_t Distributions_t + NAV_T}{\sum_t Contributions_t}, \tag{1}$$

$$IRR = \text{internal rate of return of fund cash flows}, \tag{2}$$

$$PME = \frac{\sum_t Distributions_t \cdot R_{m,t}}{\sum_t Contributions_t \cdot R_{m,t}}, \tag{3}$$

where $R_{m,t}$ denotes the public-market return used to compound cash flows.

Interpretation: PME asks whether investing the same cash-flow schedule in the public market would have performed better or worse than the private fund.

2.5 GPME and stochastic discount factors

The generalized public market equivalent uses a stochastic discount factor (SDF) to discount fund cash flows. For the market-risk model, the SDF is

$$M_{t+1} = \exp(a - b r_{m,t+1}), \tag{4}$$

where $r_{m,t+1}$ is the log market return.

The standard PME corresponds to a log-utility SDF with

$$a = 0, \qquad b = 1. \tag{5}$$

The GPME estimates a and b from public-market returns and risk-free rates.

Interpretation: GPME adjusts PME for risk exposure. If a private fund has high market beta during rising public markets, PME can overstate performance relative to GPME.

2.6 PME wedge

The key statistic is the PME wedge:

$$Wedge = PME - GPME. \tag{6}$$

- A positive wedge means PME is higher than GPME.
- In the Korteweg–Nagel logic, the wedge is informative about market risk exposure.
- A larger wedge implies greater market beta when public equity markets are rising.

Interpretation: The paper turns a distortion in PME into an identification tool: the size and slope of the PME wedge reveal relative risk exposure.

3 Models and empirical specifications

3.1 Artificial leverage test

The paper artificially levers fund cash flows to study how the PME wedge changes with leverage. Conceptually,

$$L_{i,t+h}(k) = C_{i,t+h} + k\left(C_{i,t+h} - C'_{i,t+h}\right), \tag{7}$$

where k is the leverage factor and C' denotes the cash flows of the public market portfolio used to replicate the fund's cash-flow timing.

Interpretation: If the strategy has high market beta, increasing artificial leverage should make the PME-GPME wedge rise more steeply.

3.2 Long-short portfolio test

The paper constructs vintage-year portfolios that are long in benchmark funds and short in impact funds:

$$LS_t = CashFlow_t^{Benchmark} - CashFlow_t^{Impact}. \tag{8}$$

Interpretation: If matched funds and impact funds had the same beta, the long-short portfolio should have near-zero market exposure and a flat PME wedge. A rising wedge implies the benchmark has higher beta than impact.

3.3 Alternative factor models

The market-only SDF is extended to multiple factors:

$$M_{t+1} = \exp(a - b' r_{t+1}), \tag{9}$$

where r_{t+1} can include:

- the public market factor;
- a public sustainability index factor;
- an emerging markets index factor.

Interpretation: The financial merit of impact investing depends on the relevant investor wealth portfolio. A sustainability investor or emerging-market investor may price risk differently from a market-index investor.

3.4 Performance interpretation

The paper reports PME and GPME as performance per dollar of capital committed. A negative GPME means the fund loses value relative to the SDF-implied risk-adjusted public benchmark:

$$GPME < 0 \quad \Rightarrow \quad \text{risk-adjusted underperformance.} \tag{10}$$

Interpretation: The central question is not only whether impact funds earn lower total returns, but whether they earn lower returns after accounting for the risks they bear.

4 Identification and empirical design

4.1 What identifies risk exposure

- The authors exploit the difference between PME and GPME for the same fund cash flows.
- They compare impact funds with matched non-impact funds and VC funds.
- They examine how the PME wedge changes under artificial leverage.
- They construct long-short portfolios to compare impact and benchmark risk exposure directly.

Interpretation: The identification is cash-flow based. It avoids relying on NAV-based returns, which are often smoothed, stale, and subject to valuation discretion.

4.2 Why matching matters

- Impact funds differ from other private-market funds in vintage, asset class, size, geography, and mission focus.
- Matching on asset class, vintage, and size reduces confounding from general private-market trends.
- The VC benchmark provides a broader comparison to the asset class most similar to many impact funds.

Interpretation: The paper is careful not to compare impact funds mechanically to all private equity. It uses comparable strategies wherever possible.

4.3 Why the long-short test matters

- If impact and benchmark funds had the same market beta, a long benchmark–short impact portfolio should have near-zero beta.
- The paper finds a positive and increasing wedge for long-short portfolios.
- This implies the benchmark side is more exposed to market risk.

Interpretation: The long-short evidence is a direct test that impact funds are less cyclical than matched and VC funds.

4.4 Limitations

- The impact sample is relatively small and young.
- Cash-flow data are private and difficult to validate outside the IFRC/Preqin data-building process.
- The strategy-level beta is inferred from cash flows rather than estimated from frequent traded returns.
- Alternative risk models are informative but cannot fully exhaust all investor wealth-portfolio possibilities.

Interpretation: The paper's strength is not a large public-market panel. Its strength is a new cash-flow dataset and a method that is robust to private-market return measurement problems.

5 Tables and figures

The figures and tables below are directly cropped from the paper and grouped compactly.

5.1 Mission, geography, industries, and asset classes

Read:

- Table 1 shows that impact funds have environmental, social, and combined social/environmental missions.
- The modal broad mission is social; many funds also pursue combined goals.
- Figure 1 shows that impact funds are much more exposed to emerging-market regions than matched or VC funds.
- Figure 2 shows impact funds are concentrated in areas such as business services, manufacturing, health, and energy, with industry distribution different from benchmarks.
- Figures 3 and 4 show that venture capital is the most common asset class and that many funds are first or second funds in a sequence.

Economic message: The sample is not a generic VC sample. Impact funds differ in geography, industry, and mission, which is why risk adjustment and careful benchmarks are necessary.

ission focus for sample impact funds.
e collect information on mission focus from a combination of sustainability reports, ospectuses, and websites. In Panel A, we provide the breakdown across social, ıvironmental, and combination social/environmental mission focus. In Panel B, we ovide further detail on the primary mission focus of the fund. Funds often have ultiple goals; for example, funds whose primary mission is economic development ten have sustainable practices or diversity, equity and inclusion (DEI) as a secondary cus. For this table, we assign each fund to a single primary focus. Primary mission cus labels are defined as follows: economic development refers to a primary objective improving economic conditions and employment in underserved regions; sustainable actices incorporate both environmental and social considerations in the sourcing ıd governance of portfolio companies; clean technology and energy funds invest imarily in clean technology and renewable energy companies; social services funds vest primarily in sectors that provide social goods to underserved populations, ch as healthcare, education, or real estate; other environment includes sustainable frastructure, water, and more general green innovation; DEI focuses on supporting omen and underrepresented minorities; and food and agriculture focuses on food curity and sustainable agriculture.

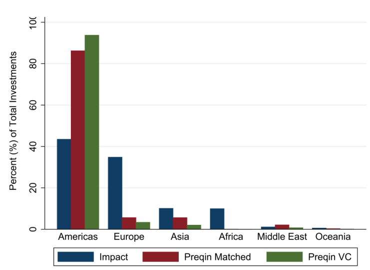
Panel A: Mission category

	Num. funds
Environmental	18
Social	48
Social and environmental	28
Total	94

Panel B: Primary mission focus

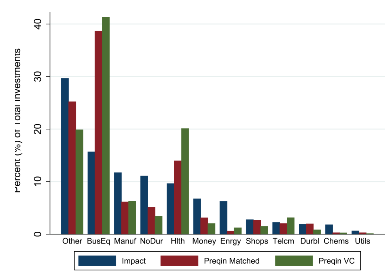
	Num. funds
Clean technology and renewable energy	14
Diversity, equity, and inclusion	4
Economic development	31
Financial inclusion and microfinance	9
Food and agriculture	2
Other environment	5
Social services	8
Sustainable practices	21
Total	94

(a) Table 1: mission focus.



g. 1. Impact and Benchmark Fund Geographic Focus.
e plot the share of investments by company headquarter region for impact, matched, ıd VC funds. This figure plots the count of investments per region as a share of total

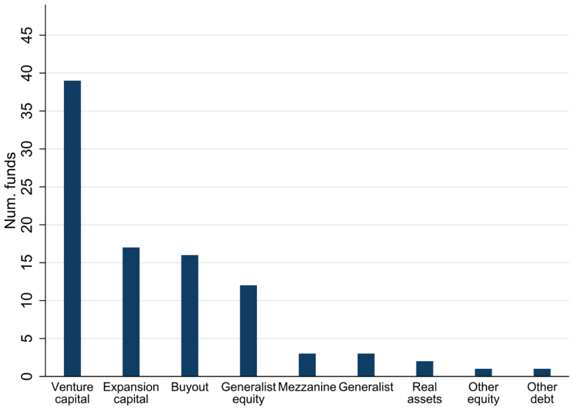
(b) Figure 1: geography.



g. 2. Impact and Benchmark Fund Industries.
e plot the share of investments by industry for impact, matched, and VC funds. ıis figure is created by counting the number of times one of the Fama–French 12

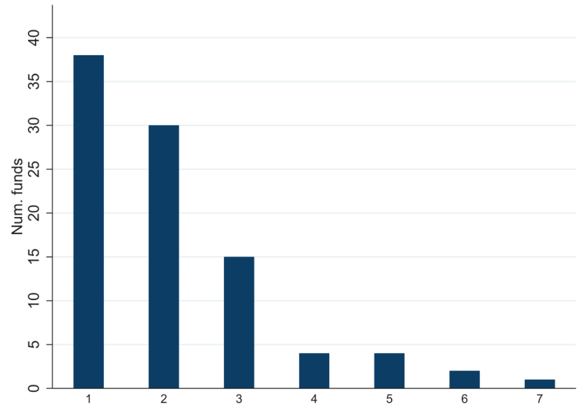
(c) Figure 2: industries.

vest with a variety of styles in companies at various stages.



g. 3. Impact Funds by Asset Class.
e plot the number of impact funds in our sample by asset class. VC funds are equity nds that invest with an early stage focus. Other equity funds include late stage and ore generalist funds. Buyout funds are equity funds with a buyout focus that use

Figure 3: impact funds by asset class



g. 4. Impact Funds by Sequence Number.
e plot the number of impact funds in our sample by sequence number, i.e. whether e fund is first, second, or further along in a series.

Figure 4: impact funds by sequence number

5.2 Table 2: fund summary statistics

Read:

- Impact funds have average vintage 2009.1 and median vintage 2009.5.
- Impact fund size averages about $429 million, with a median of about $143 million.
- The average impact KS PME is 0.743, below the matched-fund average of 0.994 and the VC average of 0.914.
- The average impact IRR is negative, while matched and VC funds have positive average IRRs.
- The cash-flow timing and number of contributions/distributions differ somewhat across samples.

Economic message: Impact funds appear to underperform on total-return measures, but these raw measures do not yet adjust for risk exposure.

Table 2
Summary statistics.
Impact fund data comes from the IFD and Preqin. VC and matched fund data come from Preqin, where matched funds are paired to impact funds using asset class, vintage, and size. All samples cover vintages from 1999 through 2015 with transaction dates from 1999 to 2021. Impact fund statistics are reported in columns 1–3, matched fund statistics are in columns 4–6, and VC fund statistics are in columns 7–9. We report the mean and median for each sample of funds. The vintage year is the year of fund inception and fund size is the total committed capital raised by the fund, reported in millions USD. The KS PME is the Kaplan and Schoar (2005) ratio described in Equation A.1, where R_{mt} is the returns on the S&P 500 total return index:

$$PME_{KS} = \frac{\sum_t \frac{distribution_t}{1+R_{mt}}}{\sum_t \frac{contribution_t}{1+R_{mt}}}$$

We calculate the KS PME using quarterly data when available and annual data for the remaining impact funds. VC and matched PMEs are all calculated using quarterly data. We report other absolute measures of performance, the cash flow multiple and IRR in percentage points. All three absolute performance measures are winsorized by 2.5% on both sides. Finally, we report characteristics about the cash flow profile of each set of funds. Effective years is the number of years funds in each sample are operating. We also report the number of total cash flows per fund, as well as the number of contributions and distributions separately.

	Impact			Matched			VC		
	N	Mean	Median	N	Mean	Median	N	Mean	Median
Vintage	94	2009.1	2009.5	94	2009.1	2009.5	484	2006.1	2006
Fund Size (Mill $)	94	428.9	143.2	94	419.8	160	484	372.7	280
KS PME	94	0.743	0.731	94	0.994	0.923	484	0.914	0.823
Multiple	94	1.154	1.083	94	1.529	1.335	484	1.568	1.327
IRR (%)	94	−0.325	3.395	94	9.935	9.037	480	5.374	5.185
Effective Years	94	7.8	7.7	94	7.8	7.8	484	11.9	12.4
# Cash Flows per Fund	94	25.6	25.5	94	29.2	27	484	33.8	32
# Contributions	94	17.3	17	94	17.8	17	484	20	19
# Distributions	94	8.3	6	94	11.4	10	484	13.8	12

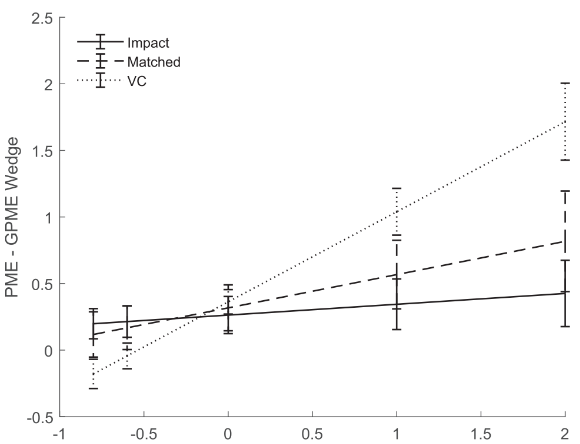
5.3 Market risk: Tables 3–5 and Figures 5–6

Read:

- Table 3 reports the market SDF parameters used to compute GPME.
- Figure 5 shows that the PME-GPME wedge grows much more steeply for VC and matched funds than for impact funds.

- Table 4 shows that impact funds have PME of −0.206 and GPME of −0.470 relative to market risk.
- Matched funds have GPME of −0.330 and VC funds have GPME of −0.433.
- Table 5 shows long-short portfolios. Long matched–short impact has GPME near zero, while long VC–short impact has negative GPME.
- Figure 6 shows that long-short portfolios have a positive and increasing PME-GPME wedge, implying benchmarks have higher market beta than impact.

Economic message: Impact funds lose money relative to the public market after market risk adjustment, but the difference with comparable private-market funds is small. The risk profile is clearly different: impact funds are less exposed to market risk.

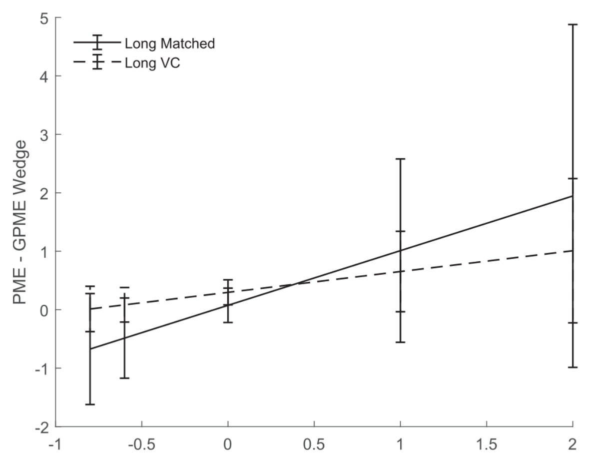


g. 5. PME-GPME Wedge Using Market Risk SDF.
e plot the PME-GPME wedge of artificially levered funds with different leverage k.
e create artificially levered fund cash flows $L_{i,t+h(j)}$ using impact funds, VC funds, or
nds matched to our impact sample on asset class, vintage, and size, $C_{i,t+h(j)}$, as well
the corresponding T-bill funds, $C_{if,t+h(j)}$:

$$_{,t+h(j)} = C_{i,t+h(j)} + k(C_{i,t+h(j)} - C_{if,t+h(j)})$$

ıe SDF for the GPME is the market risk SDF from Table 3, and the SDF for the PME
the log-utility CAPM SDF from the same table. The wedge is the difference between
ИE and GPME point estimates. The error bars are 95% confidence intervals using

Figure 5: market-risk PME-GPME wedge



g. 6. PME-GPME Wedge of Value-Weighted Long-Short Portfolio.
e create two sets of vintage portfolios, one that is long VC funds and short impact
nds, and the other that is long matched funds and short impact funds, where matched
nds are paired to impact funds using asset class, vintage, and size. Both portfolios
esented here are value-weighted. We create artificially levered cash flows $L_{i,t+h(j)}$ of
ıch strategy similar to Fig. 5, using the cash flows from long-short vintage portfolios,
$_{,t+h(j)}$, as well as the corresponding T-bill funds, $C_{if,t+h(j)}$:

$$_{,t+h(j)} = C_{i,t+h(j)} + k(C_{i,t+h(j)} - C_{if,t+h(j)})$$

ɔr each level of artificial leverage k, we plot the difference between the PME
ıd GPME point estimate of the long-short portfolio. The error bars are 95% confi-

Figure 6: long-short wedge

Table 3
Estimated SDF with market risk.
The SDF for both PME and GPME relative to market risk is given by $M^*_{t+1} = \exp(a - br_{t+1})$, where r_{t+1} is the log of the return on the market. The SDF corresponding to the PME is the SDF for a log-utility model, where $a = 0$ and $b = 1$. The SDF corresponding to the GPME relaxes this assumption. We use the CRSP value-weighted index on Kenneth French website as the market. We estimate this SDF using public asset portfolios to replicate pooled VC, matched, and impact fund cash flows, as described in Appendix A.

	PME SDF	GPME SDF
a	0	0.199
		(0.066)
b	1	3.913
		(0.719)

(a) Table 3: market SDF.

Table 4
Estimated performance relative to market risk.
We compute PME and GPME relative to the market risk factor using the SDF parameters from Table 3. We report the standard error in parentheses below the estimate, and the p-value of the J-test of whether the estimate is zero in brackets on the third line. Panel (a) estimates performance for impact funds, Panel (b) for nonimpact funds matched to the impact fund sample using asset class, vintage, and size, and Panel (c) for VC funds.

Panel (a): Impact funds		
	PME	GPME
Estimate	−0.206	−0.470
(sd)	(0.041)	(0.186)
[J-stat p-value]	[0.000]	[0.012]
Panel (b): Matched funds		
	PME	GPME
Estimate	−0.013	−0.330
(sd)	(0.033)	(0.184)
[J-stat p-value]	[0.707]	[0.073]
Panel (c): VC funds		
	PME	GPME
Estimate	−0.070	−0.433
(sd)	(0.088)	(0.120)
[J-stat p-value]	[0.424]	[0.000]

Table 5

(b) Table 4: market-risk performance.

Estimated performance of value-weighted long-short portfolios relative to market risk.
We compute PME and GPME relative to the market risk factor for two sets of value-weighted vintage portfolios: Panel (a) estimates performance for the portfolios that are long matched funds and short impact funds, where matched funds are paired to impact funds using asset class, vintage, and size, and Panel (b) for the portfolios that are long VC funds and short impact funds. We report the standard error in parentheses below the estimate, and the p-value of the J-test of whether the estimate is zero in brackets on the third line. Both panels use the SDF parameters from Table 3.

Panel (a): Match-impact		
	PME	GPME
Estimate	0.060	−0.015
(sd)	(0.054)	(0.099)
[J-stat p-value]	[0.265]	[0.881]
Panel (b): VC-impact		
	PME	GPME
Estimate	0.138	−0.159
(sd)	(0.079)	(0.069)
[J-stat p-value]	[0.080]	[0.022]

wedge should become negative as levered β falls below one and the

(c) Table 5: long-short performance.

5.4 Sustainability risk: Tables 6–8 and Figure 7

Read:

- Table 6 adds a public sustainability factor to the market model.
- For impact funds, GPME improves from −0.470 under the market-only model to −0.377 under the market-plus-sustainability model.
- Matched and VC funds do not benefit as much from adding the sustainability factor.
- Table 7 reports SDF parameters for a one-factor sustainability-index model.
- Table 8 shows that performance relative to the sustainability index differs from market-risk performance.
- Figure 7 shows the sustainability-index PME-GPME wedge. Impact funds have lower sustainability-index risk exposure than benchmarks.

Economic message: Sustainability risk helps explain impact fund returns, consistent with impact investing having a distinct exposure to sustainability-related public risk.

ıble 6
?ME with Multi-Factor SDF.
e compute the GPME for the one-factor model with market factor only and for e two-factor model with market and sustainability index. We report the standard ror in parentheses below the estimate, and the *p*-value of the J-test of whether the timate is zero in brackets on the third line. Panel (a) estimates performance for ıpact funds, Panel (b) for nonimpact funds matched to the impact fund sample using set class, vintage, and size, and Panel (c) for VC funds. All three panels use the same)F parameters; each column has its own set of SDF parameters corresponding to the levant model.

Panel (a): Impact funds		
	Market Factor Only	Market and Sustainability
Estimate	−0.470	−0.377
(sd)	(0.186)	(0.193)
[J-stat p-value]	[0.012]	[0.050]
Panel (b): Matched funds		
	Market Factor Only	Market and Sustainability
Estimate	−0.330	−0.341
(sd)	(0.184)	(0.188)
[J-stat p-value]	[0.073]	[0.070]
Panel (c): VC funds		
	Market Factor Only	Market and Sustainability
Estimate	−0.433	−0.397
(sd)	(0.120)	(0.121)
[J-stat p-value]	[0.000]	[0.001]

ıble 7

(a) Table 6: market plus sustainability factor.

the log of the return on the sustainability index. The SDF corresponding to the PME the SDF for a log-utility model, where $a = 0$ and $b = 1$. The SDF corresponding to the ?ME relaxes this assumption. We use the Dow Jones Sustainability World total return dex as the sustainability index. We estimate this SDF using public asset portfolios to plicate pooled VC, matched, and impact fund cash flows, as described in Appendix A.

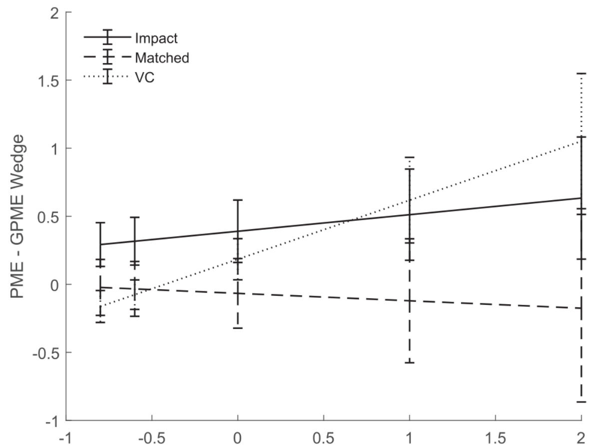
	PME SDF	GPME SDF
a	0	0.202 (0.071)
b	1	4.275 (0.948)

(b) Table 7: sustainability SDF.

Table 8
Estimated Performance Relative to Sustainability Index.
We compute the PME and GPME relative to the sustainability index using the SDF parameters from Table 7. We report the standard error in parentheses below the estimate, and the *p*-value of the J-test of whether the estimate is zero in brackets on the third line. Panel (a) estimates performance for impact funds, Panel (b) for nonimpact funds matched to the impact fund sample using asset class, vintage, and size, and Panel (c) for VC funds. All three panels use the same SDF parameters; each column has its own set of SDF parameters corresponding to the relevant model.

Panel (a): Impact funds		
	PME	GPME
Estimate	−0.123	−0.513
(sd)	(0.044)	(0.289)
[J-stat p-value]	[0.005]	[0.075]
Panel (b): Matched funds		
	PME	GPME
Estimate	0.159	0.225
(sd)	(0.040)	(0.270)
[J-stat p-value]	[0.000]	[0.405]
Panel (c): VC funds		
	PME	GPME
Estimate	0.107	−0.077
(sd)	(0.129)	(0.140)
[J-stat p-value]	[0.404]	[0.584]

(c) Table 8: sustainability-index performance.



g. 7. PME-GPME Wedge with Sustainability Index.
e plot the PME-GPME wedge of artificially levered funds with different leverage k.
e create artificially levered funds as in Fig. 5, using impact funds, VC funds, or funds
atched to our impact sample on asset class, vintage, and size, $C_{i,t+h(j)}$, as well as the
›rresponding T-bill funds, $C_{if,t+h(j)}$:

$$_{,t+h(j)} = C_{i,t+h(j)} + k(C_{i,t+h(j)} - C_{if,t+h(j)})$$

ıe SDF for the GPME is the sustainability risk SDF from Table 7, and the SDF for the
ИE is the log-utility SDF from the same table. The wedge is the difference between
ИE and GPME point estimates. The error bars are 95% confidence intervals using
›otstrap standard errors from 1000 bootstrap samples of impact or benchmark funds.

5.5 Emerging-market risk: Tables 9–10 and Figure 8

Read:

- Table 9 reports SDF parameters using an emerging markets index.
- Table 10 shows performance relative to the emerging markets index.
- Impact funds have near-zero GPME relative to emerging markets, while matched and VC funds have very large positive GPMEs with large standard errors.
- Figure 8 shows that matched and VC funds have sharply different emerging-market wedge patterns from impact funds.

Economic message: The relevant wealth portfolio matters. For investors already exposed to emerging markets, the relative financial merit of impact investing can look different than for market-index investors.

Table 9
Estimated SDF with emerging markets index.
The SDF relative to the emerging markets index is given by $M^*_{t+1} = \exp(a - br_{t+1})$, where r_{t+1} is the log of the return on the emerging markets index. The SDF corresponding to the PME is the SDF for a log-utility model, where $a = 0$ and $b = 1$. The SDF corresponding to the GPME relaxes this assumption. We use the MSCI Emerging Markets Investable Market total return index as the emerging markets index. We estimate this SDF using public asset portfolios to replicate pooled VC, matched, and impact fund cash flows, as described in Appendix A.

	PME SDF	GPME SDF
a	0	0.402
		(0.081)
b	1	6.229
		(1.120)

(a) Table 9: EME SDF.

Estimated performance relative to emerging markets index.
We compute the PME and GPME relative to the Emerging Markets Index using the SDF parameters from Table 9. We report the standard error in parentheses below the estimate, and the p-value of the J-test of whether the estimate is zero in brackets on the third line. Panel (a) estimates performance for impact funds, Panel (b) for nonimpact funds matched to the impact fund sample using asset class, vintage, and size, and Panel (c) for VC funds. All three panels use the same SDF parameters; each column has its own set of SDF parameters corresponding to the relevant model.

Panel (a): Impact funds

	PME	GPME
Estimate	–0.068	–0.018
(sd)	(0.061)	(4.233)
[J-stat p-value]	[0.270]	[0.997]

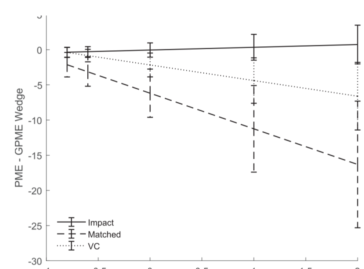
Panel (b): Matched funds

	PME	GPME
Estimate	0.271	6.458
(sd)	(0.075)	(6.010)
[J-stat p-value]	[0.000]	[0.283]

Panel (c): VC funds

	PME	GPME
Estimate	0.172	2.343
(sd)	(0.196)	(2.348)
[J-stat p-value]	[0.380]	[0.318]

(b) Table 10: EME performance.



g. 8. PME-GPME Wedge with Emerging Markets Index.
e plot the PME-GPME wedge of artificially levered funds with different leverage k.
e create artificially levered funds as in Fig. 5, using impact funds, VC funds, or funds
atched to our impact sample on asset class, vintage, and size, $C_{i,t+h(j)}$, as well as the
›rresponding T-bill funds, $C_{if,t+h(j)}$:

$$_{,t+h(j)} = C_{i,t+h(j)} + k(C_{i,t+h(j)} - C_{if,t+h(j)})$$

ıe SDF for the GPME is the emerging markets risk SDF from Table 9, and the SDF
r the PME is the log-utility SDF from the same table. The wedge is the difference
:tween PME and GPME point estimates. The error bars are 95% confidence intervals
ing bootstrap standard errors from 1000 bootstrap samples of impact or benchmark
nds.

(c) Figure 8: EME wedge.

6 Short summary

- The paper builds a new cash-flow dataset for private impact investing funds.
- Impact funds have lower raw performance than matched private-market and VC funds.
- The paper uses PME, GPME, and the PME-GPME wedge to infer risk exposure from fund cash flows.
- Impact funds have lower market risk exposure than matched and VC funds.
- After market risk adjustment, impact underperforms the public market, but it is not necessarily worse than comparable private-market strategies.
- Long-short portfolios confirm that matched and VC funds are more market-risk exposed than impact funds.
- Adding a sustainability factor improves the pricing of impact returns, suggesting impact has distinctive sustainability exposure.
- Using an emerging-market wealth portfolio changes the relative assessment of impact performance.
- The perceived financial merit of impact investing depends on investor preferences, risk exposures, and the benchmark wealth portfolio.

7 Conclusion

7.1 Core conclusion

Jeffers, Lyu, and Posenau show that impact investing in private markets has a distinct risk profile. Impact funds have lower market risk exposure than comparable private-market strategies. They underperform the public market after market risk adjustment, but not necessarily more than matched private-market funds.

7.2 Economic intuition

The core intuition is that lower absolute returns do not automatically imply concessionary risk-adjusted returns. If impact funds are less cyclical or hedge risks that investors care about, their returns should be evaluated relative to that risk exposure. The PME-GPME wedge provides a cash-flow-based way to infer this exposure.

7.3 Takeaway

The paper reframes the debate on impact investing. The question is not only whether impact funds earn high returns, but what risks they bear and which investor portfolio is the relevant benchmark. Impact investing can be financially unattractive for a market-index investor while still playing a useful role for investors who value lower market risk, sustainability exposure, or impact itself.